Angular Momentum as Information Geometry: Isotropic Packing and the Recurring Angular Cross-Section

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[**SYMBOLIC PROOF**] We show that the angular momentum label set (ℓ, m) is counted exactly by a geometric construction: uniform-density packing of distinguishable states on concentric isotropic shells in two dimensions. Starting from two axioms—discrete binary distinguishability and geometric indifference (maximum entropy)—and the construction's structural choices (motivated, not forced), the construction yields per-shell angular capacities $2k-1$ for shell k, equal in count to the spherical harmonic basis $Y_{\ell m}$ with $\ell = k-1$ and $m = -\ell, \ldots, +\ell$ (assigning the m values to positions is a labelling choice). [**OBSERVATION**] The principal quantum number n corresponds to a cumulative grouping of the first n shells, producing the n^2 orbital count, but this grouping is not forced by the axioms alone. Similarly, the per-shell count carries a factor of 2 from the binary initialization ($N_{\text{init}} = 2$), which we read as corresponding to spin multiplicity (a counting correspondence, not a derivation of spin). On the label set taken as a graph, the Laplacian splits into ℓ-blocks (the edges change n or m, never ℓ) and so does not converge to the S^3 Laplacian—the spinless node set is the exact S^3 label set, on which $-\Delta_{S^3} = L^2 + A^2$ is the diagonal $SO(4)$ Casimir n^2-1—and [**INTERNAL THEOREM**] its top eigenvalue saturates the bipartite bound $\lambda_{\max} \to 2d_{\max} = 8$ (attained in a mid-ℓ block, not in the s-wave block, whose own bound is 4), so that under a single matched scale $K_{\text{vac}} = -1/16$ the extremal eigenvalue reaches the hydrogen ground-state energy—matched, not derived: the content that could have failed is the spectral bound, not the target the scale is matched to. The per-shell spectrum $E_n = -Z^2/(2n^2)$ is not a function of the graph Laplacian. On the labels it is the function $E_n = -Z^2/(2(C+1))$ of the diagonal Casimir $C = n^2 - 1$, i.e. the bound-state Hamiltonian is $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$, and that function is fixed by Fock's continuum projection: the eigenvalue $2\pi^2/n$ of the S^3 Coulomb kernel forces the focal length $p_0 = Z/n$, and the energy shell $p_0^2 = -2E$ then gives E_n (Paper 7). [**OBSERVATION**] This angular cross-section recurs across the natural geometries of the Geometric Vacuum series as a basis label: the full (ℓ, m) labeling, with $2\ell+1$ states per ℓ, labels the basis on S^3 for hydrogen, the per-electron partial waves of the hyperspherical treatment of helium, the angular channels of the molecule-frame hyperspherical treatment, and each fiber of the composed bundles (ℓ is a *conserved* quantum number only for the one-electron atom; elsewhere it is a channel label coupled by Gaunt coefficients); in the two-centre prolate-spheroidal geometry only m survives as a label, because the η-separation replaces ℓ by a separation index that reduces to it only in the united-atom limit. This is Paper 0 in the series.

INTRODUCTION

The hydrogen atom possesses n^2 orbital states per principal quantum number n, each labeled by angular momentum $\ell = 0, 1, \ldots, n-1$ and magnetic quantum number $m = -\ell, \ldots, +\ell$, with a spin multiplicity of 2 giving $2n^2$ states total. The standard derivation obtains these labels from the eigenvalue structure of the angular momentum operators $\hat{L}^2$ and $\hat{L}_z$ acting on the Hilbert space of the Schrödinger equation. Here we ask: to what extent can this quantum number structure be recovered from geometry alone, without invoking operators, Hilbert spaces, or the eigenvalue equation?

We present a construction based on uniform-density packing of distinguishable states on concentric isotropic shells in two dimensions. The construction rests on two axioms: (i) a minimum of two distinguishable states is required to define a metric (binary distinguishability), and (ii) in the absence of external constraints, the distribution of states must maximize entropy, enforcing isotropy (geometric indifference). These axioms, together with the natural choices of concentric shells and area-proportional state counting, produce the following results at three distinct levels of strength.

[**SYMBOLIC PROOF**] **Strong claim: angular quantum numbers.** Shell k has capacity $2k-1$, which equals in count the spherical harmonic basis $Y_{\ell m}$ with $\ell = k-1$: the per-shell degeneracy $2\ell+1$ is an exact count identity, and attaching the labels (ℓ, m) to the positions is a labelling choice. This count identity is the central result of the paper.

[**OBSERVATION**] **Structural correspondences.** The principal quantum number n arises when one groups the first n shells cumulatively, yielding the count $\sum_{k=1}^{n}(2k-1) = n^2$. This grouping is natural but not uniquely forced by the axioms. The factor of 2 in the per-shell count comes from the binary initialization ($N_{\text{init}} = 2$); the compactified packing manifold S^2 is also orientable (two orientations, a $\mathbb{Z}_2$), which offers a second, heuristic reading of the same factor. [**OBSERVATION**] The identification of this factor of 2 with spin-$\frac{1}{2}$

multiplicity is a counting correspondence, not a derivation from the axioms.

[OBSERVATION] Series significance. The (ℓ, m) angular sector counted here recurs inside the natural geometries of the Geometric Vacuum (GeoVac) series as a basis label. The full (ℓ, m) labeling, with $2\ell + 1$ states per ℓ, appears on S^3 for hydrogen [10] (where ℓ is also conserved), in the per-electron angular factors of the hyperspherical coordinates for He [13], in the angular channels of the molecule-frame hyperspherical treatment of H_2 [14], and in each fiber of the composed bundles for LiH [15]; in the multi-electron and two-centre cases ℓ is a channel label coupled by Gaunt coefficients, not a conserved quantum number. In the prolate spheroidal coordinates for H_2^+ and H_2 [11, 12] only m is a label: the η-separation replaces ℓ by a separation index whose node count correlates with ℓ only in the united-atom limit. Where it appears, the full geometry separates into a system-specific radial or hyperradial part and an angular cross-section whose labeling, fixed by the rotational symmetry the geometry retains, is the one the packing counts. This recurrence is the reason the packing construction is foundational to the series rather than merely a curiosity about hydrogen.

Several elements of this work connect to well-established results. The counting $\sum_{\ell=0}^{n-1}(2\ell + 1) = n^2$ is standard SO(3) representation theory. The relationship between hydrogen's "accidental" degeneracy and the geometry of S^3 was established by Fock [1] and Bargmann [2]. What is new here is the specific packing algorithm that reproduces the per-shell count of the (ℓ, m) labels from geometric–combinatorial reasoning independent of the operator formalism, and the observation that this angular sector recurs as the angular fiber of the GeoVac natural geometries wherever their symmetry retains it (fully at the one-centre, molecule-frame hyperspherical and composed levels; only through m at the two-centre prolate level).

The paper is organized as follows. Section presents the packing algorithm. Section establishes the count identity between shell capacity and the spherical harmonic basis. Section discusses the structural correspondences for n and spin. Section shows how the angular cross-section recurs across the GeoVac natural geometries. Section constructs the graph topology and states what its spectrum does and does not carry. Sections and address scope, limitations, and context.

THE PACKING CONSTRUCTION

Axioms

The construction rests on two axioms, which contain no reference to quantum mechanics, operators, wavefunctions, or physical observables.

Axiom 1 (Binary Distinguishability). To define a metric space, a minimum of two distinguishable states ($N_{\text{init}} = 2$) is required to establish a non-zero fundamental scale d_0. A single point cannot define a distance; two points create the first bit of spatial information.

Note: The choice $N_{\text{init}} = 2$ is the minimal non-trivial initialization. It is the simplest choice consistent with this axiom, but not the only one; initializing with $N_{\text{init}} = 3$ or more points would define a different fundamental area and modify the downstream counting.

Axiom 2 (Geometric Indifference). In the absence of external constraints or privileged directions, the distribution of states must maximize entropy. By the Principle of Indifference, this necessitates isotropy: states populate isotropic shells (circles) rather than anisotropic tilings (squares, hexagons), which would imply privileged directions not present in the axioms.

Note: Isotropy motivates circular shells over anisotropic tilings, but uniform-density packing on concentric circles is one of several ways to implement isotropy—for instance, hexagonal close-packing achieves higher packing density while preserving six-fold symmetry. Concentric shells with area-proportional counting is the simplest realization consistent with the axioms, not the only one.

These axioms, together with the specific choices of concentric circular shells, area-proportional counting, and binary initialization, constitute the full set of assumptions. The axioms motivate these choices but do not uniquely determine them. What is remarkable is that this particular realization—arguably the simplest one consistent with the axioms—reproduces exactly the per-shell count of the angular momentum quantum numbers.

Algorithm

Step 1. Place $N_{\text{init}} = 2$ points on an isotropic shell of radius $r_1 = d_0$, separated by the fundamental distance d_0.

Rationale: Two points are the minimum needed to define a distance (Axiom 1). Isotropy requires a circular shell (Axiom 2). This fixes both d_0 and the fundamental area per state $\sigma_0 = \pi d_0^2/2$.

Step 2. Add concentric isotropic shell k at radius $r_k = k \cdot d_0$.

Step 3. Determine the state count N_k on shell k from uniform information density. The annular area between shells $k - 1$ and k is

$$A_k = \pi r_k^2 - \pi r_{k-1}^2 = \pi d_0^2\big[k^2 - (k-1)^2\big] = \pi d_0^2(2k-1)\,. \tag{1}$$

Dividing by the fundamental area gives the state count:

$$N_k = \frac{A_k}{\sigma_0} = \frac{\pi d_0^2(2k-1)}{\pi d_0^2/2} = 2(2k-1)\,. \tag{2}$$

Step 4. Repeat for $k = 1, 2, 3, \ldots$

Resulting structure

The construction yields the distribution shown in Table I.

Shell k	States $N_k = 2(2k-1)$	Cumulative $\sum_{i=1}^{k} N_i$
1	2	2
2	6	8
3	10	18
4	14	32
5	18	50
⋮	⋮	⋮
k	$2(2k-1)$	$2k^2$

TABLE I. State distribution on isotropic shells. Each shell k carries $2(2k-1)$ states; the cumulative count through shell k is $2k^2$.

Two features of this table merit immediate comment. First, the per-shell count $2(2k-1)$ factorizes as $2 \times (2k-1)$, where the factor $2k-1$ is an odd integer determined entirely by the area formula. Second, the cumulative count $2k^2$ matches the hydrogen degeneracy $2n^2$ if one identifies $k = n$. These observations are analyzed in the next two sections at appropriately different levels of strength.

THE ANGULAR COUNT UNDER UNIFORM DENSITY

This section contains the central result of the paper: the per-shell state count from the packing construction equals in count the angular momentum labeling (ℓ, m).

The angular capacity of shell k

The factor of $2(2k-1)$ in Eq. (2) factorizes into a global factor of 2—whose origin in the initialization condition (and a heuristic orientability reading) is analyzed in Sec. —and an angular capacity per shell:

$$N_k^{\text{angular}} = 2k - 1 \tag{3}$$

distinct angular slots. **[SYMBOLIC PROOF]** The annular-area derivation of this $2k-1$ angular capacity (equivalently $2\ell+1$ with $\ell = k-1$) is verified by `tests/test_trunk_qa_annular.py`.

Count identity with spherical harmonics

The spherical harmonics $Y_{\ell m}(\theta, \phi)$ for a given ℓ have $2\ell+1$ independent components labeled by $m = -\ell, -\ell+1, \ldots, +\ell$. Under the identification

$$\ell = k - 1\,, \tag{4}$$

the angular capacity of shell k becomes

$$N_k^{\text{angular}} = 2(k-1) + 1 = 2\ell + 1\,, \tag{5}$$

which is precisely the dimension of the spin-ℓ irreducible representation of SO(3). The $2k-1$ angular positions on shell k are in one-to-one correspondence with the magnetic quantum numbers $m = -\ell, \ldots, +\ell$.

[SYMBOLIC PROOF] The equality of counts need not be checked case by case: it is an algebraic identity. The area formula (Eq. 1) forces the annular area of shell k to be proportional to $2k-1$, which is the dimension of the $(k-1)$-th irreducible representation of SO(3). (The identity establishes equal counts for every k; it is not a mechanism linking planar area to SO(3), and the bijection of positions with m values is a labelling choice.) The packing construction therefore reproduces, shell by shell, the count of the complete angular momentum label set $\{(\ell, m) : 0 \le \ell,\ -\ell \le m \le \ell\}$.

What the construction produces

To be precise about the scope of this result:

- The construction reproduces the *degeneracy* $2\ell+1$ per shell as an exact count; attaching the angular labels (ℓ, m) to its positions is a labelling choice.
- It does *not* produce the functional form of the spherical harmonics, the eigenvalue $\ell(\ell+1)$ of $\hat{L}^2$, or the commutation relations of angular momentum operators. Those require additional structure: the operators themselves (the $SO(3)$ Casimir L^2, whose eigenvalue is $\ell(\ell+1)$, and the $so(3)$ commutation relations) and, for the functional forms, the continuum harmonics reached through Fock's projection (Papers 1 and 7). A diagonal Casimir on the labels supplies eigenvalues, not functional forms or commutation relations; nor does a continuum limit of the graph Laplacian supply them, since it does not converge to Δ_{S^3} (Sec.).
- The result is *independent* of any specific physical system. The angular count comes from 2D isotropic geometry, not from the hydrogen atom. Hydrogen merely happens to use this same angular sector—as do all rotationally symmetric quantum systems.

STRUCTURAL CORRESPONDENCES: n, SPIN, AND $2n^2$

The per-shell count of the angular quantum numbers (ℓ, m) follows directly from the packing (Sec.). The principal quantum number n and the spin factor of 2 involve additional identifications that are natural and mathematically grounded but not uniquely forced by the two axioms. We present them here as structural correspondences.

The principal quantum number as cumulative grouping

If one groups the first n shells cumulatively, the total angular state count is

$$\sum_{k=1}^{n}(2k-1) = n^2\,, \tag{6}$$

which matches the orbital degeneracy of hydrogen's n-th principal shell: $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$. Including the spin factor gives $2n^2$.

This grouping is natural—the shells are concentrically ordered, so cumulative counting is the simplest aggregation—but the axioms do not *require* that shells be grouped this way. The packing generates an infinite sequence of shells; the decision to collect the first n of them into a single principal shell is an additional structural choice that corresponds to, but is not derived from, the axioms.

The factor of 2 and orientability

The per-shell count $N_k = 2(2k-1)$ contains a factor of 2 that traces back to two sources:

1. **Initialization.** Axiom 1 places $N_{\text{init}} = 2$ states on the first shell, fixing the fundamental area $\sigma_0 = \pi d_0^2/2$. This factor of 2 propagates through all subsequent shells via the uniform-density condition.

2. **Orientability (a heuristic reading).** When the flat packing plane is conformally compactified to the Riemann sphere $S^2 \cong \mathbb{C} \cup \{\infty\} = \mathbb{CP}^1$, the resulting manifold is orientable: it admits two orientations (inward and outward normals), a $\mathbb{Z}_2$ that can be read as a doubling of the state space. This $\mathbb{Z}_2$ is the orientation class, relating O(3) to SO(3); it is *not* the spin double cover SU(2)→SO(3), which comes from $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$. The arithmetic factor of 2 is the initialization above; orientability supplies no additional factor.

[**OBSERVATION**] The identification of this factor of 2 with the spin-$\frac{1}{2}$ degree of freedom is a counting correspondence: the initialization doubling and the spinor representation of SU(2) both produce a factor of 2. However, the packing construction does not derive the spin angular momentum algebra $[\hat{S}_i, \hat{S}_j] = i\epsilon_{ijk}\hat{S}_k$ or the eigenvalue $s(s+1)$. The correspondence is between the *counting* (a factor of 2) and spin *multiplicity* ($2s+1 = 2$ for $s = 1/2$), not between the construction and the full spinor formalism.

A remark on the axioms

The construction follows from the two axioms together with several additional structural choices: concentric shells with linearly spaced radii ($r_k = kd_0$), area-proportional state counting, and the specific fundamental area $\sigma_0 = \pi d_0^2/2$ fixed by the initialization. These choices are natural and mutually consistent—they are arguably the simplest possibilities compatible with the axioms—but they are not uniquely determined by the axioms alone. The axioms constrain the construction; they do not uniquely fix every detail of it.

This distinction matters for the epistemic status of the results. The per-shell count of the angular quantum numbers (ℓ, m) follows from any isotropic, uniform-density, concentric construction with linearly spaced radii—a robust class (the linear spacing is needed for the $2k-1$ per-shell capacity; other spacings change it). The specific identification of n and the factor of 2 depend on the particular choices above.

THE PACKING AS RECURRING ANGULAR CROSS-SECTION

The packing construction of Sec. counts a recurring (ℓ, m) angular sector. In the subsequent papers of the GeoVac series, this angular sector appears as a structural factor inside the natural geometries used for quantum chemistry calculations, as a basis label: the full (ℓ, m) labeling, with $2\ell+1$ states per ℓ, at Levels 1, 3 and 4 and in each Level-5 fiber (ℓ conserved only at Level 1, a Gaunt-coupled channel label elsewhere), but only m at the two-centre Level 2, where the prolate separation replaces ℓ. Where it appears, the full geometry of each system separates into a system-specific radial (or hyperradial) coordinate and the angular cross-section counted here.

Table II summarizes the hierarchy.

Level 1: Hydrogen on S^3. Fock's 1935 stereographic projection [1] maps the hydrogen momentum-space problem onto the three-sphere S^3. The eigenstates of the Laplace–Beltrami operator on S^3 are hyperspherical harmonics, which decompose as $Y_{n\ell m} =$

Level	System	Full Geometry	Angular Cross-Section	Ref.
1	H (1-center, 1e)	S^3 (Fock projection)	$Y_{\ell m}$ on $S^2 \subset S^3$	[10]
2	H_2^+, H_2 (2-center, 1–2e)	Prolate spheroidal	m exact; ℓ-like index from η-separation	[11, 12]
3	He (1-center, 2e)	Hyperspherical	$Y_{\ell m}$ per electron in fiber	[13]
4	H_2 (2-center, 2e)*	Mol-frame hyperspherical	$Y_{\ell m}$ angular channels	[14]
5	LiH (core + valence)	Composed fiber bundle	$Y_{\ell m}$ per electron group	[15]

TABLE II. **[OBSERVATION]** The (ℓ, m) angular cross-section inside each natural geometry of the GeoVac series. At Levels 1, 3, 4 and in each Level-5 fiber the coordinate system separates into a system-specific radial part and the full (ℓ, m) labeling counted in this paper (a basis label; conserved only at Level 1); at the two-centre Level 2 only m is a label. *H_2's natural geometry is Level 2 (prolate spheroidal, Paper 12); Level 4 is its structural N-electron hyperspherical treatment.

$R_{n\ell}(\chi)\, Y_{\ell m}(\theta, \phi)$ (the radial factor depends on ℓ, $R_{n\ell} \propto \sin^\ell \chi\, C^{\ell+1}_{n-\ell-1}(\cos\chi)$). The angular factor $Y_{\ell m}$ carries the per-shell count $2\ell + 1$ that the packing reproduces, with $\ell = k - 1$. The graph's node set is exactly the label set of these harmonics (the Peter–Weyl truncation of $L^2(S^3)$), on which $-\Delta_{S^3} = L^2 + A^2$ acts as the diagonal $SO(4)$ Casimir $n^2 - 1$; Paper 7 [10] verifies Fock's conformal chain from that continuum operator onward (under it the hydrogen bound-state Hamiltonian is the function $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$ of that operator, its scale fixed by the eigenvalue $2\pi^2/n$ of Fock's S^3 Coulomb kernel; that non-definitional step is tested for $n \leq 4$, general n being Fock's). The graph's edge Laplacian does *not* converge to it (Sec.).

Level 2: H_2^+ and H_2 on prolate spheroids. For diatomics, the natural geometry is the prolate spheroidal coordinate system (ξ, η, ϕ), in which the one-electron Schrödinger equation separates; it is also H_2's natural geometry, where a prolate-spheroidal CI with exact algebraic integrals recovers **[MEASURED]** 99.97% of the H_2 dissociation energy (0.053 mHa) with explicit r_{12} correlation, and 99.81% (0.32 mHa) with a re-based orbital CI (Paper 12 [12]). The azimuthal quantum number m survives exactly as the quantum number of the ϕ-equation ($e^{im\phi}$). The quantum number ℓ is no longer exactly conserved in the two-center problem: it is replaced by a separation parameter from the η-equation that reduces to ℓ in the united-atom limit. Of the (ℓ, m) labels, m is exact, but ℓ is no longer a label: the prolate η-separation replaces it by a separation index whose node count correlates with ℓ only in the united-atom limit (the lifting of the $(2\ell + 1)$-fold degeneracy by the axial potential is common to every two-centre problem, Level 4 included; what is specific to Level 2 is that the separated coordinates carry no ℓ label at all). Paper 11 [11] constructs a discrete lattice in these coordinates.

Level 3: Helium in hyperspherical coordinates. For two-electron atoms, the six-dimensional configuration space is parameterized by a hyperradius R and five angles. The angular eigenstates decompose into single-electron angular momenta (ℓ_1, m_1) and (ℓ_2, m_2), each carrying the per-shell structure of the packing. Paper 13 [13] implements this as a fiber bundle: the base space is the hyperradius, and the fiber is the angular manifold.

Level 4: H_2 in molecule-frame hyperspherical coordinates. The structural N-electron treatment of two-electron diatomics combines the molecular axis (internuclear separation R) with the hyperspherical treatment of the two electrons. The angular channels are labeled by per-electron (ℓ, m) channel labels, coupled by the two-centre potential, that determine the multichannel expansion. **[MEASURED]** Paper 15 [14] recovers 96.0% of the H_2 dissociation energy using this geometry with a Schwartz cusp correction; H_2's natural geometry, and its best result, is the prolate Level 2 above.

Level 5: LiH via composed geometries. For core–valence systems, the total geometry is a fiber bundle $G_{\text{total}} = G_{\text{nuc}} \rtimes G_{\text{core}}(R) \rtimes G_{\text{val}}(R)$, where each electron group occupies its own natural geometry. The angular labeling (ℓ, m) appears independently in each fiber. Paper 17 [15] applies this to LiH and BeH^+.

[OBSERVATION] The recurrence of the angular sector across the five levels follows from the rotational symmetry each geometry retains, and it is not uniform: m is an exact label wherever axial symmetry survives; ℓ is an exact *basis* label (with $2\ell+1$ states per ℓ) wherever the separated coordinates carry an angular S^2 factor—Levels 1, 3, 4 and each Level-5 fiber—but a *conserved* quantum number only where the full SO(3) survives (Level 1); at Level 2 the prolate separation replaces ℓ by a separation index. The packing construction counts this angular factor.

The node structure as combinatorial invariant

The preceding survey reveals a structural feature that merits explicit comment. **[OBSERVATION]** The discrete quantum-number set $\{(\ell, m) : 0 \leq \ell,\ -\ell \leq m \leq \ell\}$—with its degeneracy $2\ell + 1$ and selection rules—appears inside several continuum geometries, as the labels of the rotational (SO(3), or axial SO(2)) symmetry each retains. The five geometries catalogued in Table II differ in their separated coordinates and coordinate singularity structures. S^3 is compact with constant positive curvature; prolate spheroidal coordinates are a coordinate system on flat $\mathbb{R}^3$ with coordinate singularities at the foci; the hyperspherical manifold S^5 is compact but five-dimensional (for two electrons); the composed fiber bundle is a product of manifolds with different dimensionalities at each level. Levels 1, 3 and 4, and each fiber of Level 5, share the full (ℓ, m) node structure as a basis labeling, with its $2\ell + 1$ count and angular momentum coupling rules (ℓ conserved only at Level 1); the two-

centre Level 2 shares only the m label, its η-separation index reducing to ℓ only in the united-atom limit.

This invites a comparison with a standard universality argument familiar from statistical mechanics and lattice field theory, where many different lattice discretizations (square, triangular, hexagonal) flow under renormalization to the same continuum field theory, and the continuum description is usually read as fundamental. Here the shared object runs the other way: one discrete label structure (fully, or through m alone at Level 2) is carried by several continuum geometries. This does not make either side prior: the label set and the continuum operators are dual descriptions of the same quantum-number structure, and the observation is about what they share, not about which is fundamental.

We state this as a mathematical observation, not a metaphysical claim. The conformal equivalences between each continuum operator and standard quantum mechanics are not established uniformly across the hierarchy. For the S^3 embedding the lattice's labels are the operator's eigenbasis labels, the hydrogen bound-state Hamiltonian is the function $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$ of that operator (Fock), and **[SYMBOLIC PROOF]** Paper 7 [10] provides 18 symbolic checks of Fock's continuum projection chain (ten of them limiting-case, definitional or non-discriminating) (a graph-to-continuum *operator* convergence is not part of it: the edge Laplacian does not converge to Δ_{S^3}, Sec.). At the other levels only the angular channel structure carries the packing labels (at Level 2 the prolate separation constants are not labelled by ℓ at all, only by m and a node count), and the subsequent papers benchmark each geometry in the hierarchy numerically. **[OBSERVATION]** Levels 1–4 are the standard continuum problems in their separated coordinates; the composed Level 5 adds a Phillips–Kleinman pseudopotential and is an approximation to, not an equivalent of, the all-electron problem. Whether one regards the discrete or continuum description as more fundamental is a choice of interpretation, not a consequence of the mathematics.

An analogy to isospectral manifolds is apt. Kac asked whether one can hear the shape of a drum [18]; Milnor [19] had already exhibited isospectral, non-isometric flat tori, and Gordon, Webb and Wolpert [20] answered Kac's question negatively for planar domains. **[OBSERVATION]** The present result differs in a specific sense: the geometries in Table II share a labeled node structure—quantum number assignments and per-shell state counts, with the (ℓ, m) selection rules shared as basis labels (m in every row; ℓ in every row except the two-center Level 2, where the prolate separation replaces it). The spectra themselves differ across rows; what is shared is combinatorial structure, which is richer than a list of eigenvalues and different in kind from isospectrality.

GRAPH TOPOLOGY AND WHAT ITS SPECTRUM CARRIES

Vertex set

We define a graph $G = (V, E)$ on the label set (reached from the packing's shells only through the cumulative-grouping Observation, which reuses shell k in every principal shell $n \geq k$), with vertex set

$$V = \{(n, \ell, m, s) : 1 \leq n \leq n_{\max},\ 0 \leq \ell < n,\ -\ell \leq m \leq \ell,\ s \in \{\uparrow, \downarrow\}\} \tag{7}$$

with cardinality $|V| = \sum_{n=1}^{n_{\max}} 2n^2 = \frac{1}{3}\, n_{\max}(n_{\max} + 1)(2n_{\max} + 1)$ for a lattice truncated at $n_{\max}$: each shell n contributes its $2n^2$ spin-doubled capacity, summed over $1 \leq n \leq n_{\max}$. (The top shell alone holds $2n_{\max}^2$; the cumulative count is the sum.) **[SYMBOLIC PROOF]** This spin-doubled vertex count is verified for all $n_{\max}$ symbolically (`tests/test_paper0_vertex_count.py`) and by enumeration (`tests/test_dirac_lattice.py`, which works in the (n, κ, m_j) labelling; the spinless production lattice holds $\sum n^2$ nodes).

Edge set

Edges encode geometrically adjacent transitions:

- **Radial edges** ($\hat{T}_\pm$): connect (n, ℓ, m, s) to $(n \pm 1, \ell, m, s)$ at fixed angular quantum numbers (ℓ, m). They connect labels related by the ladder operators $\hat{T}_\pm$; they are not physical transitions.
- **Angular edges** ($\hat{L}_\pm$): connect (n, ℓ, m, s) to $(n, \ell, m \pm 1, s)$ at fixed n and ℓ. They connect labels related by the ladder operators $\hat{L}_\pm$ within a shell; they are not physical transitions.

The full graph topology may also include edges connecting states with different ℓ (e.g., ℓ-changing transitions driven by non-spherical perturbations). The production graph has no ℓ-changing edges, so its Laplacian decomposes into independent ℓ-blocks (m is mixed within a block by the azimuthal edges). This block-diagonality mirrors the ℓ-conservation of the spherically symmetric Coulomb Hamiltonian, but it is a property of the chosen edge set, not a consequence of the graph encoding the Coulomb problem; it is also what prevents the graph Laplacian from converging to Δ_{S^3} (below).

Spectral bound

The graph Laplacian $L = D - A$ defines the legacy graph Hamiltonian (not the physical hydrogen Hamiltonian)

$$H = K_{\text{vac}}\, L\,, \tag{8}$$

where $K_{\rm vac} = -1/16$ is the kinetic scale factor. It matches only the graph's extremal eigenvalue (a band edge, $\lambda_{\max} \to 8$) to the hydrogen ground-state energy $-1/2$; it does not carry the rest of the graph spectrum onto the Rydberg ladder, whose per-shell energies come from Fock's continuum projection: the S^3 Coulomb kernel's eigenvalue $2\pi^2/n$ forces $p_0 = Z/n$, and the energy shell $p_0^2 = -2E$ then gives E_n (Paper 7; the label Casimir $C = n^2 - 1$ alone does not fix the function, but once Fock's kernel fixes it, $E_n = -Z^2/(2(C+1))$ on the labels). **[OBSERVATION]** This constant's value 1/16 *coincides* with a conformal-geometry quantity of the Fock projection (Paper 7 [10]): the squared coupling between adjacent n-shells in the Gegenbauer eigenbasis of the S^3 Laplacian is $c^2(n,l) = \frac{1}{4}[1-l(l+1)/(n(n+1))]$ (Paper 7), and the quantity that coincides with 1/16 is the fourth power $\Omega(0)^{-4}$ of the inverse Fock conformal factor, with $\Omega(0) = 2$, equivalently $c^2(n,0)/4$. This numerical coincidence between the energy-matching scale and the geometric 1/16 is an observation, not a derivation (Paper 7): the matching scale and the conformal quantity are computed from disjoint inputs, with no known mechanism connecting them.

[INTERNAL THEOREM] The edges change n (radial, $T_\pm$) or m (azimuthal, $L_\pm$) but never ℓ, so the Laplacian splits into ℓ-blocks; the $\ell = 0$ block is a path graph in n whose top eigenvalue tends to 4, while the full graph's largest Laplacian eigenvalue converges to $\lambda_{\max} \to 8 = 2d_{\max}$, twice the coordination number, attained in a mid-ℓ block ($\ell = 11$ at $n_{\max} = 30$) (`tests/test_trunk_qa_kappa.py`, `tests/test_paper7_graph_convergence.py`). **[INTERNAL THEOREM]** The saturation has a closed-form rate. Because no edge changes ℓ, the Laplacian is block diagonal in ℓ and each block is the grid graph $P_{n_{\max}-\ell} \times P_{2\ell+1}$ (radial × azimuthal paths), so the whole spectrum is **[PRIOR ART]** the textbook Laplacian spectrum of a Cartesian product of path graphs [6] (the block identification and the constant $C_{\rm sat}$ below are this paper's)

$$\mathrm{spec}(L) = \Big\{\, 2-2\cos\frac{j\pi}{n_{\max}-\ell} \;+\; 2-2\cos\frac{k\pi}{2\ell+1} \,\Big\}, \quad (9)$$

Each block's top eigenvalue is $4+2\cos\frac{\pi}{n_{\max}-\ell}+2\cos\frac{\pi}{2\ell+1}$, so the deficit is $\min_\ell\big[4\sin^2\frac{\pi}{2(n_{\max}-\ell)} + 4\sin^2\frac{\pi}{2(2\ell+1)}\big]$; minimising its leading form $\pi^2[(n_{\max}-\ell)^{-2}+(2\ell+1)^{-2}]$ under the block constraint gives the constant in closed form,

$$2d_{\max}-\lambda_{\max} \;=\; \frac{C_{\rm sat}}{n_{\max}^2}+o\big(n_{\max}^{-2}\big), \qquad C_{\rm sat} \;=\; \frac{\pi^2}{4}\big(2+2^{1/3}\big)^2\big(1+2^{-2/3}\big) \;=\; 42.7397\ldots, \quad (10)$$

attained at $\ell/n_{\max} \to 1 - 2/(2+2^{1/3}) = 0.386488$ (measured argmax $\ell = 11$ at $n_{\max} = 30$, 27 at 70). **[MEASURED]** Every sample checked lies *below* $C_{\rm sat}$ (all 591 integer cutoffs $10 \le n_{\max} \le 600$) — $C_{{\rm sat},n} =$ 42.61 at $n_{\max} = 320$, 42.70 at 1000, 42.73 at 4000 — so every sample checked understates $C_{\rm sat}$ (a measured one-sided pattern, not proved for all $n_{\max}$). The approach is *not* monotone, however: there are 195 decreasing steps in that range, e.g. $C_{{\rm sat},20} = 40.7285 > C_{{\rm sat},21} = 40.6470$. It is the one-sided pattern, not monotonicity, that makes every checked sample an understatement (the 591-cutoff scan, the $C_{{\rm sat},20} > C_{{\rm sat},21}$ step and the $n_{\max} = 320$ and 1000 values are pinned by `tests/test_lattice_spectrum.py`; the closed form by `tests/test_paper1_block_spectrum.py`). Finally, with $H = K_{\rm vac}L$ and $K_{\rm vac}$ fixed by the match $K_{\rm vac} = E_{\rm target}/8 = -0.5/8$, the ground-state eigenvalue $K_{\rm vac}\lambda_{\max}$ of H tends to -0.5 Ha as $\lambda_{\max} \to 8$, by construction. The convergence of the spectral bound is the content; the reproduction of the target is not. **[WALL]** Nor is that bound an S^3 quantity: $8 = 2d_{\max}$ is the band edge of any large two-dimensional grid, and $C_{\rm sat}$ is a Dirichlet band edge. The load-bearing obstruction is the kernel dimension: because the ℓ-blocks are disconnected, the Laplacian on the spinless (n,ℓ,m) node set has $\dim\ker L = n_{\max}$ at every cutoff, whereas Δ_{S^3} has a single zero mode, so L does *not* converge to Δ_{S^3}, for the binary production weights and the $1/(n_i n_j)$ option alike (`tests/test_graph_s3_nonconvergence.py`). (Relatedly, the graph heat kernel e^{-tL} vanishes identically between blocks; turning that into a statement about the continuum heat kernel would need a node-to-point identification this paper does not make, so it is corroboration, not the argument.) The spin-doubled vertex set of Eq. (7) is two copies of this node set; the kernel and label statements here are made on the spinless set. What is exact is the spinless node set: it is the Gel'fand–Tsetlin label set of the Peter–Weyl truncation of $L^2(S^3)$, on which $-\Delta_{S^3} = L^2 + A^2$ acts as the diagonal $SO(4)$ Casimir n^2-1 at every cutoff (a node weight, not an edge Laplacian). The **[SYMBOLIC PROOF]** (ℓ, m) label set, with its $2\ell+1$ count, is exact at every finite $n_{\max}$—it is carried by the node labels, not approximated by mesh refinement.

The conformal equivalence between the hydrogen problem and the Laplace–Beltrami operator on the unit S^3 (via Fock's stereographic projection, under which the bound-state Hamiltonian is the function $H = -\frac{Z^2}{2}(1-\Delta_{S^3})^{-1}$ of that operator) is developed in Paper 7 [10], **[SYMBOLIC PROOF]** where an audit of 18 symbolic checks (ten of them limiting-case, definitional or non-discriminating) validates the conformal geometry of that continuum chain, and an exact kernel-eigenvalue test carries its non-definitional energy step, $p_0 = Z/n$ (the graph Laplacian is not a discretization of it; see above). Paper 1 [17] develops the operator algebra on these labels (the $SU(1,1) \times SU(2)$ ladder operators, exact on interior shells, with the Rydberg energies imposed through the energy shell, $p_0 = Z/n$ fixed by Fock's kernel eigenvalue (Paper 7), rather than output by the graph); dynamics

were attempted via the graph Laplacian in the archived Paper 6 [21] (superseded).

SCOPE AND LIMITATIONS

What the construction provides

- ✓ The per-shell degeneracy $2\ell+1$ of the angular labels (ℓ, m), reproduced as an exact count by uniform-density isotropic packing.
- ✓ A graph topology (V, E) defined on the labels, whose edges connect labels related by ladder operators (not physical transitions).
- ✓ The structural correspondences $n \leftrightarrow$ cumulative grouping and spin multiplicity $\leftrightarrow$ the initialization factor of 2 (a counting correspondence), with their proper epistemic status identified.

What the construction does not provide

- × Energy eigenvalues $E_n = -1/(2n^2)$. These come from Fock's continuum projection (the eigenvalue $2\pi^2/n$ of the S^3 Coulomb kernel forces $p_0 = Z/n$ on the shell where the $SO(4)$ Casimir is n^2-1, and the energy shell $p_0^2 = -2E$ converts it to E_n; Paper 7), not from the graph Laplacian, whose spectrum lies in $[0, 8]$; the matched kinetic scale K_{vac} fixes only the extremal eigenvalue (its value is matched, and its coincidence with the Fock-Jacobian factor $1/16$ is an Observation rather than a derivation; Papers 1, 7).
- × Wavefunctions $R_{n\ell}(r)$ or $Y_{\ell m}(\theta, \phi)$ in their functional form. The packing counts the labels; it does not produce the functions.
- × Time evolution and dynamics. These require the Hamiltonian as an operator; they were attempted via the graph Laplacian in the archived Paper 6 [21] (superseded).
- × The fine structure constant or any coupling constants. The construction is purely kinematic.

Choices beyond the axioms

As noted in Sec. , the construction involves structural choices—concentric shells, linearly spaced radii ($r_k = kd_0$), area-proportional counting, and the fundamental area $\sigma_0 = \pi d_0^2/2$—that are natural and mutually consistent but not uniquely determined by the two axioms. **[OBSERVATION]** The robustness of the (ℓ, m) result rests on the observation that any isotropic, uniform-density, concentric packing with linearly spaced radii ($r_k = kd_0$, so the k-th annulus has area proportional to $k^2 - (k-1)^2 = 2k-1$) produces the angular capacity $2k-1$ per shell; the linear spacing is required (proved just below) — other radial spacings change the per-shell capacity. **[SYMBOLIC PROOF]** Conversely, capacity $2k-1$ for every k with $r_0 = 0$ forces $r_k = kd_0$: the condition $r_k^2 - r_{k-1}^2 = (2k-1)d_0^2$ telescopes to $r_k^2 = \sum_{j\le k}(2j-1)\, d_0^2 = k^2 d_0^2$ (`tests/test_trunk_qa_annular.py`).

DISCUSSION

Historical context

The connection between geometry and quantum mechanics has been recognized since the earliest days of quantum theory. Bohr [9] imposed quantization as a geometric constraint (angular momentum $L = n\hbar$ on circular orbits, later reinterpreted by de Broglie [5] as an integer number of wavelengths around the orbit). Fock [1] discovered that the hydrogen momentum-space problem maps onto S^3 via stereographic projection, revealing the SO(4) dynamical symmetry that explains the "accidental" n^2 degeneracy. Bargmann [2] derived the same symmetry algebraically. Malkin and Man'ko [3] and Fronsdal [4] introduced the conformal group $SO(4,2)$ as hydrogen's dynamical group, and Barut and Kleinert [7] extended the symmetry analysis to transition probabilities via noncompact dynamical groups. Biedenharn and Louck [8] provided a comprehensive treatment of the angular momentum algebra underlying the (ℓ, m) structure.

The present work operates at a different level: rather than analyzing the symmetries of the Schrödinger equation, it reproduces the angular count from geometric–combinatorial reasoning that is logically independent of the operator formalism. The result has the same per-shell count as the (ℓ, m) basis, obtained by a different route.

The kinematic–dynamic distinction

The packing construction reproduces the angular degeneracy counting of the *kinematic* structure of quantum mechanics. It does not produce the *dynamics*: the Hamiltonian, the energy spectrum, or time evolution. This separation is a feature of the construction, not a limitation. It shows that, given the construction's choices (Sec.), the per-shell count of the angular quantum-number labels follows from geometry and counting, independently of the specific force law or potential.

In the GeoVac series, dynamics enter through the Coulomb problem's continuum structure, not through a limit of the graph Laplacian. Paper 7 [10] verifies Fock's

conformal equivalence between the hydrogen problem and the Laplace–Beltrami operator on S^3 (the bound-state Hamiltonian being $H = -\frac{Z^2}{2}(1 - \Delta_{S^3})^{-1}$), under which the $1/r$ Coulomb potential can equivalently be described as the coordinate distortion of the stereographic projection (a dual description, not a claim that either reading is prior); the graph's (spinless) node set is the label set of that operator's harmonics, on which $-\Delta_{S^3} = L^2 + A^2$ acts as the diagonal $SO(4)$ Casimir $n^2 - 1$. The graph Laplacian equipped with $K_{\text{vac}} = -1/16$ does *not* converge to that operator: it splits into $n_{\max}$ disconnected ℓ-blocks (§VI).

Scope of the geometric analogy

The construction shows that information-theoretic reasoning under geometric constraints *can* reproduce the per-shell count of the angular labels. It does not show that this is the *only* way to obtain these labels, nor does it establish that quantum mechanics *must* have a geometric origin. The angular momentum algebra $[\hat{L}_i, \hat{L}_j] = i\epsilon_{ijk}\hat{L}_k$ is a consequence of rotational symmetry regardless of whether one arrives at it through operator theory or geometric packing. What the construction demonstrates is that the number $2k - 1$ has a geometric–combinatorial derivation; its equality with the degeneracy $2\ell + 1$ is a count identity, not a derivation of the degeneracy.

CONCLUSIONS

The per-shell count of the angular momentum quantum numbers (ℓ, m) follows from uniform-density packing on isotropic shells: shell k carries $2k - 1$ angular slots, equal in count to the spherical harmonic basis $Y_{\ell m}$ with $\ell = k-1$. This result follows from two axioms (binary distinguishability and geometric indifference) together with natural structural choices (concentric shells with linearly spaced radii $r_k = k\,d_0$, uniform density; conversely, capacity $2k-1$ for every k, with $r_0 = 0$, forces that spacing). It is independent of the Schrödinger equation, operator postulates, or any specific physical system.

The principal quantum number n corresponds to cumulative grouping of the first n shells, yielding n^2 orbital states. The factor of 2 in the per-shell count, set by the binary initialization, corresponds in counting to spin multiplicity. These are structural correspondences—genuine mathematical observations that match hydrogen's quantum numbers—but they depend on identifications beyond what the two axioms alone determine.

The (ℓ, m) angular sector counted here recurs as the angular cross-section inside the natural geometries of the GeoVac series as a basis label—in full on S^3 for single-electron atoms [10] (where ℓ is also conserved), as Gaunt-coupled channel labels in the hyperspherical and molecule-frame hyperspherical treatments and in each fiber of the composed bundles for core–valence molecules [15]; only through m in the two-centre prolate geometry, where the separation replaces ℓ. The computational framework of the series is built on this label set, on Fock's continuum projection, and on the natural geometries in which each problem separates—not on the graph Laplacian and its kinetic scale, whose spectrum is a two-dimensional grid band structure and does not carry the hydrogen ladder (§VI). It has been applied to H, He, H_2^+, H_2, HeH^+, LiH, and BeH^+, benchmarked against exact and experimental values. **[MEASURED]** H_2 reaches 99.97% of D_e (0.053 mHa) in its natural prolate geometry with explicit r_{12} correlation (Paper 12 [12]); in that same prolate engine the HeH^+ equilibrium bond length reaches -0.5% at $l_{\max} = 3$ (from -4.5% at $l_{\max} = 2$; Paper 12); the LiH equilibrium geometry is $+0.2\%$ in an all-electron prolate treatment with an analytic core (Paper 19 [16]) and 5.3% in the composed Level 5 treatment (Paper 17), where BeH^+ is markedly worse (R_{eq} 3.25 vs $\sim$2.48 bohr with the ab-initio pseudopotential); and in the mol-frame hyperspherical treatment (Paper 15 [14]) HeH^+ at fixed bond length recovers 93% of D_e adiabatically but only $\sim$5% in the consistent 2D-variational solver. These inherited figures are owned, and test-backed or not, by the cited papers.

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